

Economic History II

Seasonality, agriculture, and the origins of the state

FGV EPGE

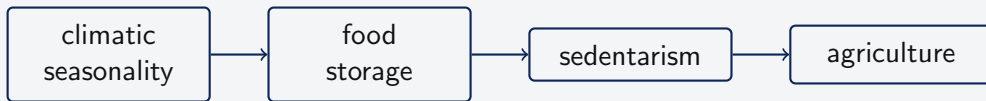
The puzzle

- The long-run gains from agriculture are obvious: more food, more people, more specialized labor.
- But the first farmers were **shorter** and had more joint diseases than hunter-gatherers.
- Read as: they ate less and worked harder.
- Even so, at least seven unconnected populations invented farming, roughly 12,000 to 5,000 years ago.
- Why accept what looked like a lower standard of living?

Today's paper

- Andrea Matranga, *The Ant and the Grasshopper: Seasonality and the Invention of Agriculture*, QJE 2024.
- Trigger: a spike in **climatic seasonality**, from well understood oscillations in the shape of Earth's orbit.
- Response: store the nonperishable wild foods you gather, to get through the season of scarcity.
- Storing means staying near the granary, so bands became sedentary before they could farm.
- Sedentarism plus storage made cultivation easy to develop.

The mechanism in one line

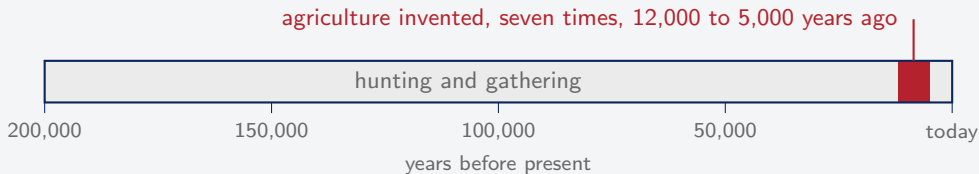


- Farming does not have to raise average consumption to be adopted.
- Eliminating the regular famines caused by dry or cold periods is enough.
- The technology arrives after an earlier change in the way of life.

What has to be explained

- Global: independent inventions in the Fertile Crescent, the Sahel, north and south China, the Andes, Mexico, North America.
- Concentrated in time, after some 200,000 years of hunting and gathering.
- Concentrated in space: temperate mid-latitudes, and monsoonal zones on either side of the Equator.
- And a biological cost: shorter skeletons, worn teeth, joint disease and infection.

Modern humans and the timing of agriculture



- Roughly 200,000 years of hunting and gathering.
- Seven independent inventions, all inside the last 12,000 years.
- The marked band is 3.5 percent of the bar.

Why the year became uneven

- Three orbital parameters: the **axial tilt**, the orbital eccentricity, and the precession of the equinoxes.
- At the peak of the last ice age, 22,000 years ago, a smaller tilt meant milder seasonal variation.
- By 12,000 BP, the Northern Hemisphere had seasonal sunlight variation unprecedented in the past 50,000 years.
- The same change ended the ice age: warm summers melted ice, and global temperature rose 7 to 8 degrees.
- So the ice age and the Neolithic are not rival explanations; they share a cause.

Temperature Difference and Invention

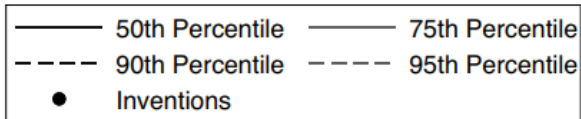
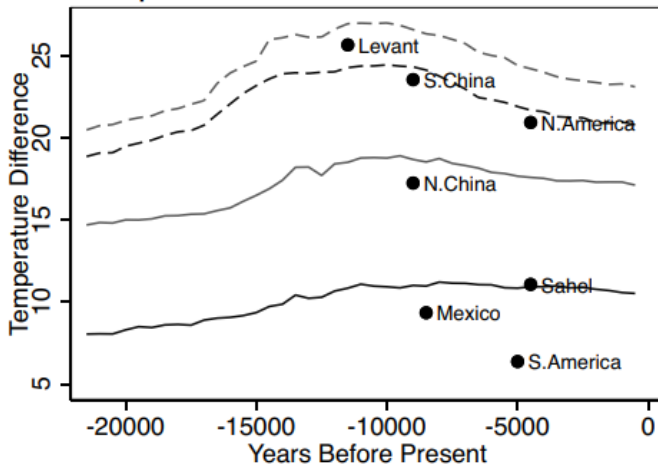


Figure 1: invention

- Vertical axis: average temperature of the warmest quarter minus that of the coldest, each censored at zero.
- Below freezing, plants do not grow, so a winter of -10 and one of -50 are the same for food.
- The curves are percentiles of that measure across the world's land cells, period by period.
- Dots: the seven independent inventions, placed at their own seasonality.
- The Levant, south China and North America sit at or above the 90th percentile of the world; north China above the 75th.
- The Sahel, Mexico and South America sit low: monsoon zones, where the seasonality is in rainfall.

Climate from TraCE-21ka, a simulation of the last 22,000 years driven by Earth's orbit, the extent of glaciation, greenhouse gases and sea level. It gives temperature and rainfall by trimester for cells of 3.75 degrees, averaged here into 44 periods of 500 years.

How seasonality is measured

- Temperature: censor each quarter at 0°C first, then take the difference.

$$\text{TempSeas} = \max(\text{Temp.Warmest}, 0) - \max(\text{Temp.Coldest}, 0)$$

- Why censor: no photosynthesis once the ground freezes, so a swing from −10 to −50 looks extreme but leaves no food either way.
- This is close to the seasonal gap in growing degree days, the standard agronomic measure.
- Rainfall: wettest month minus driest, divided by mean rainfall.

$$\text{PrecipSeas} = \frac{\text{Precip.Wettest} - \text{Precip.Driest}}{\text{MeanPrecip}}$$

- Relative, so that a place with 1,000mm in its dry season is not counted as starving.

Storage and sedentarism

What has to come before agriculture?

A band, two places, two months

	July	December
Hill	$1 + \sigma$	$1 - \sigma$
Valley	0	$1 - \sigma + \gamma$

- σ : how seasonal the climate is. γ : how much the valley adds in December.
- **Nomads** follow the food: hill in July, valley in December. They cannot store.
- **Settlers** take the hill only, and smooth by storing: one unit in each month.
- Settling costs the December refuge, so average consumption falls. It buys a flat year.

What the model says

- Mild climate (low σ) and a good December refuge (high γ): nomadism is optimal, and farming never starts.
- Whatever the refuge is worth (any γ), a large enough rise in seasonality (higher σ) makes settlement optimal.
- The more the refuge is worth (higher γ), the more seasonal the climate has to get first (higher σ).
- On settling, average consumption falls at once, because the December refuge (γ) is lost. Consumption seasonality disappears.
- It stays lower in the long run: a flat diet no longer holds fertility down, so the gain goes into population instead.

[Play with the model](#)

What counts as storage

- Not any occasional stockpile.
- The relevant case: nonperishable food, on a scale that carries the group through the lean season.
- That is what makes mobility expensive, and ties the band to the granary.
- Storage and sedentarism are therefore complements, and should appear together.
- Predictability matters: seasons repeat within the year, so how much to store is knowable. Year-to-year swings are not.

Four predictions for each sequence

1. Seasonal variability in temperature or rainfall rose before agriculture.
2. A sedentary or semisedentary phase came before agriculture.
3. Intensive storage became prominent only after populations settled.
4. Storage was practiced before agriculture.

Seven sequences, four predictions: 28 pairs to check.

	Seasonality Before Agriculture	Sedentarism Before Agriculture	Sedentarism Before or During Storage	Storage Before Agriculture	Averages
Middle East	1	1	1	1	1
Sahel	1	1	1	1	1
North China	1	1	1	1	1
South China	1	1	1	0	0.75
South America	1	1	0	1	0.75
Eastern North America	1	0	0	1	0.5
Mesoamerica	1	0	1	0	0.5
Averages	1	0.71	0.71	0.71	0.79

Figure III

- 22 of the 28 pairs are confirmed.
- The first column holds everywhere: seasonality rose before agriculture in all seven.
- Middle East, Sahel, north China: all four predictions.
- South China: no evidence of intensive storage before farming. South America: storage before full sedentarism.
- Eastern North America and Mesoamerica: half. In Mesoamerica some cultivation came first, then sedentarism and storage.

Reading the misses

- A sedentary storing phase may have been brief, and easy to miss in the record.
- Excavation and preservation are uneven across the seven regions.
- Archaeological categories do not map cleanly onto residence, extent of farming, and storage.
- Slash and burn farmers who move every few years leave remains much like foragers who move every few months.

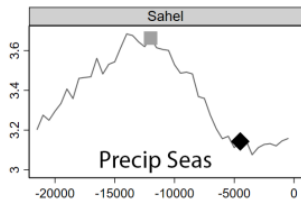
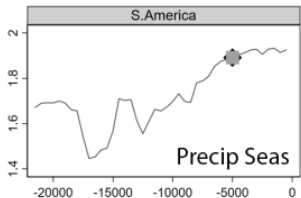
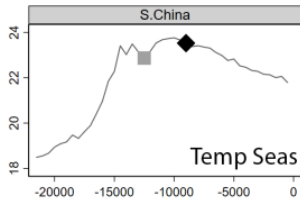
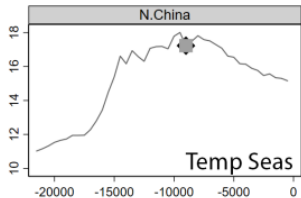
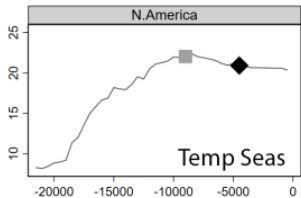
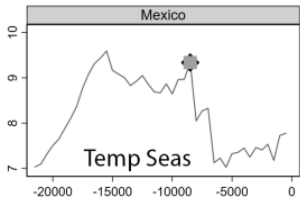
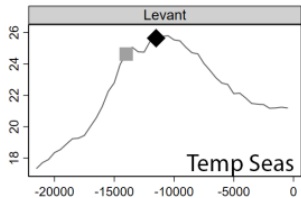


Figure A1

- One panel per center of invention, with the seasonality measure that binds there.
- Temperature seasonality for five of them; precipitation seasonality for the Sahel and South America.
- Gray square: sedentarism, where a date exists. Black diamond: agriculture.
- The seasonality peak generally precedes both markers.
- That gap is the mechanism: incentives change first, way of life second, technology third.

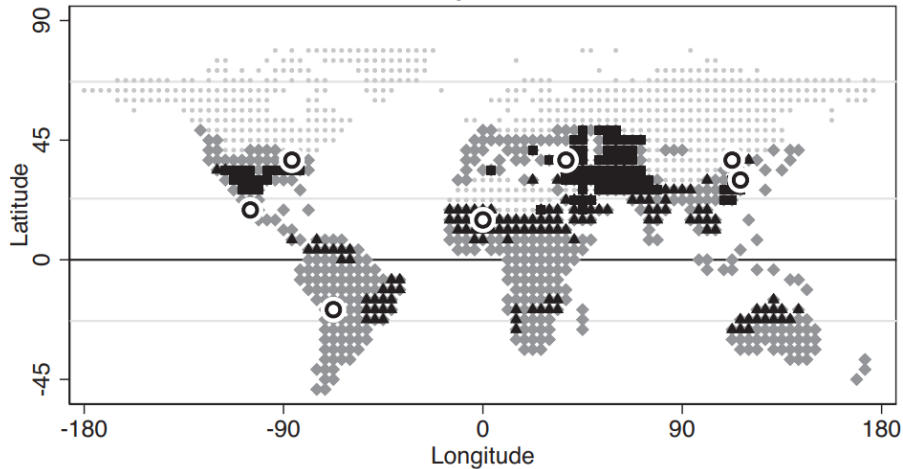
Testing the theory

From the archaeological record to a climate panel

Global data

- Climate: the TraCE simulation of the last 22,000 years, on cells of 3.75 by 3.75 degrees, collapsed into 44 periods of 500 years each.
- Adoption: Putterman and Trainor for 160 countries, dated by when the first local population drew half its calories from domesticated plants and animals.
- Invention: the seven primary domestication centers of Purugganan and Fuller, from carbon dating and crop DNA.
- 1,024 land cells; 587 warm and wet enough for farming to be plausible.
- Earliest farmers 11,500 years ago; the average qualifying location, 4,700 years ago.

12k year BP



- | | |
|------------------------------------|--------------------------------------|
| • Too Cold or Dry | ◆ Pleasant, not Seasonal |
| ■ Pleasant, High Temp. Seasonality | ▲ Pleasant, High Precip. Seasonality |
| ● Independent Invention | |

Figure IV

- A snapshot at 12,000 years before present, on the coarse climate grid.
- Squares: pleasant and highly seasonal in temperature. Triangles: pleasant and highly seasonal in rainfall.
- Circles: the seven independent inventions.
- Six of the seven sit in, or very near, areas where seasonality had risen.
- Mexico is the outlier: seasonal dry lands sit beside rain forest across a mountain range, and the grid averages them together.

When did each place start farming?

- One standard deviation more temperature seasonality, about 8 degrees between summer and winter: agriculture about **1,500 years earlier**.
- Across the full range, from a flat year to about 28 degrees: about 4,000 years earlier.
- One standard deviation more precipitation seasonality, about 0.7 index points: about 500 years earlier.
- Across its full range, up to an index of 3.5: about 2,500 years earlier.
- Holding fixed average temperature and rainfall, latitude, the Americas, and distance to the nearest center of invention.

Where was it invented?

- Each cell is followed period by period, and leaves the sample once it has agriculture, from any source.
- Chance that a place invents farming in a given 500-year window:
 - flat year: about 1 in 20,000
 - world average, 12 degrees: about 1 in 2,600
 - one standard deviation above that: about 1 in 700
 - the most seasonal places, 33 degrees: about 1 in 77
- The most seasonal places: over 30 times the average place, and 260 times a place with a flat year.
- Present-day climate added on top changes nothing, so this is not about where archaeologists dig today.

Adoption and diffusion

Once agriculture exists, who takes it up first?

Temperature Difference and Adoption

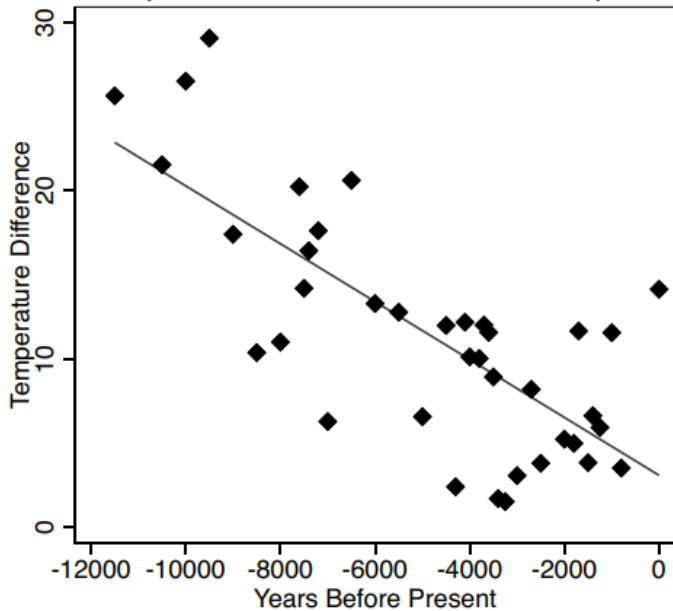


Figure 1: adoption

- Now the object is the date each place starts farming, invented locally or imported.
- Horizontal axis in years before present: further left is earlier.
- Binned averages: early adopters have a larger gap between summer and winter temperatures.
- Descriptive, and it pools invention with diffusion. The regional sample separates them.

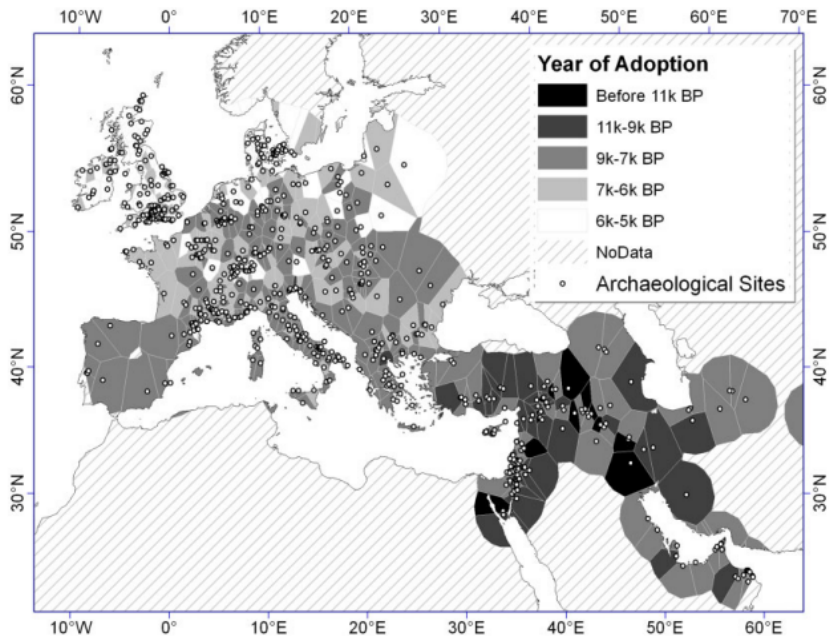


Figure V

- 750 archaeological sites in western Eurasia, each with a carbon date for the earliest definite evidence of farming.
- The earliest agriculture forms an arc from the eastern Mediterranean to the Persian Gulf.
- From the flanks of the Zagros and Taurus, out onto Mesopotamia, across the Bosphorus into the Balkans, then in two thrusts to the northern plains and the western Mediterranean.
- Average speed of diffusion: roughly one kilometer a year.
- The Fertile Crescent is fertile only relative to the desert around it. What sets it apart is being pleasant and extremely seasonal at once.

Seasonality against year-to-year volatility

- Ashraf and Michalopoulos: moderate variation *across* years helped, extremes at either end delayed.
- Matranga: predictable variation *within* the year.
- The two are hard to separate, because mid-latitudes have both. In western Eurasia they correlate at 0.6.
- Put both in together and seasonality keeps its explanatory power.
- One reading: seasonality created the preconditions, year-to-year volatility set the speed once farming had started.

The cost of agriculture

Less food on average, less seasonal hunger

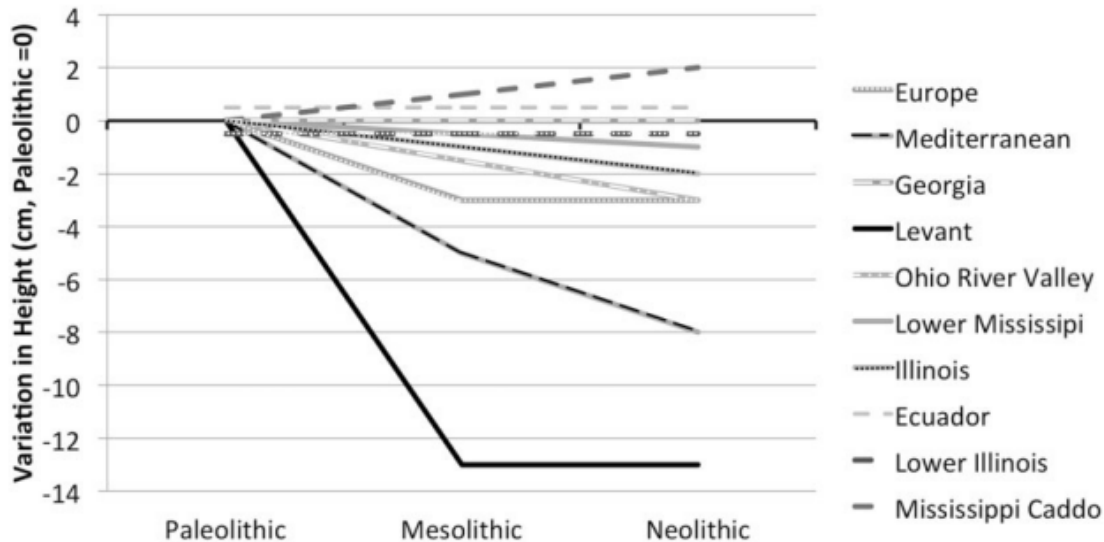


Figure VI

- Achieved adult male height by phase, one line per location, Paleolithic set at zero.
- Neolithic farmers are clearly shorter than their nomadic ancestors.
- Where Mesolithic data exist, the decline already shows up there, before farming.
- The Natufians got shorter when they settled and began storing, before cereals were a dietary mainstay.
- So the drop is not simply meat giving way to grain: late Paleolithic diets already used those plants, and early farmers still hunted.

Harris lines

- Height records average childhood nutrition, not its volatility, because catch-up growth hides episodes of starvation.
- But catch-up leaves a trace: a layer of spongy bone inside the hollow shaft, a **Harris line**.
- Cohen and Armelagos count them for seven pre- and post-transition pairs: marked decreases in five, no movement in one, a slight increase in one.
- Central Ohio Valley: nomadic hunter-gatherers averaged 165 cm with 11 lines each. After farming, about three centimeters shorter, with four lines.
- Reading: hunter-gatherers ate well on average, and starved for part of the year.

Why not the southern temperate zone?

- New Zealand, Australia, South Africa and Argentina have climates much like the places where farming began.
- None invented agriculture in the Neolithic.
- The seasonality shock happened only in the Northern Hemisphere.
- South of 30 degrees, invention was delayed even where conditions were otherwise favorable.
- Related claim: part of the tropics' late start may be delayed agriculture, since latitude and seasonality are tightly linked.

Threats to the interpretation

- Only seven inventions, so any test of invention rests on seven events.
- Adoption dates and simulated paleoclimate are both noisy, and unevenly excavated.
- Comparing each place with itself over time would leave only those seven, so it cannot be done.
- Diamond's rival story: seasonal climates breed large seeded grasses, easy to domesticate. Adding a measure of domesticable plants leaves seasonality intact.
- Comparing only within small neighborhoods, the effect shifts from temperature to rainfall.

Takeaway

- The Neolithic need not begin with a more productive technology.
- The decisive change came earlier: from mobility to sedentary storage.
- Predictable seasonality made storage worth its cost, and storage made farming easy to invent and to copy.
- It explains the shorter skeletons without anyone having made an irrevocable mistake.
- Unconnected populations, similar incentives, similar solutions, within comparable time spans.

Origins of the state

What is a state?

- Weber's definition:
 - *a human community that (successfully) claims the monopoly of the legitimate use of physical force within a given territory*
- Hobbes, *Leviathan*: everyone is roughly equal, and everyone seeks power, for pleasure or for safety.
- The corollary is anarchy, a war of all against all:
 - *continual fear, and danger of violent death; and the life of man, solitary, poor, nasty, brutish, and short*
- A power large enough to break that equality can deliver peace, and with it investment.

Olson: roving bandits

Mancur Olson, *Dictatorship, Democracy, and Development*, 1993

- A monarchist in a poor southern Italian village: “Monarchy is the best kind of government because the King is then owner of the country. Like the owner of a house, when the wiring is wrong, he fixes it”
- Small bands, 50 or 100 people, keep order by agreement: each family gets a real share of the gain.
- Large populations cannot. Each person bears the full cost of order and receives a millionth of the benefit.
- Under anarchy: uncoordinated competitive theft by roving bandits, with no stake in the future.
- Nobody produces above subsistence, so there is little left to steal.

Olson: the stationary bandit

- A bandit who seizes a domain and monopolizes theft converts plunder into regular taxation.
- His subjects keep what is left after tax, so they save and invest, and next year's take is larger.
- He now wants them alive and productive, and will supply order and other public goods when these raise taxable income.
- “the first blessing of the invisible hand”: the roving bandit settles down and replaces anarchy with government.
- “he is not like the wolf that preys on the elk, but more like the rancher who makes sure that his cattle are protected and given water”

Olson: the grasping hand

- The autocrat picks the **revenue-maximizing** tax rate, not the one that maximizes welfare.
- His appetite is not bounded by what one person can eat or wear: armies, prestige, larger domains.
- A short horizon turns him back into a roving bandit: confiscation, repudiated debts, debased coin.
- No independent court can bind him, by definition, so his promises never become fully credible.
- “History provides not even a single example of a long and uninterrupted sequence of absolute rulers who continuously respected the property and contract-enforcement rights of their subjects”

Olson: why a majority takes less

- A majority also collects taxes, but it earns market income as well.
- That makes its stake in the economy more encompassing, so its best tax rate is lower than the autocrat's.
- Narrow interests run the other way: a typical US lobby covers under 1 percent of national income-earning capacity, and ignores almost all of the social cost.
- So democracies do not necessarily redistribute less. What they lack is the autocrat's incentive to extract the maximum surplus for one person.
- What long-horizon investment needs is secure rights and predictable succession, which is what a lasting democracy supplies.

The Origin of the State: Land Productivity or Appropriability?

Joram Mayshar, Omer Moav, and Luigi Pascali



PDF



PDF PLUS



Abstract



Full Text



Supplemental Material



Abstract

The conventional theory about the origin of the state is that the adoption of farming increased land productivity, which led to the production of food surplus. This surplus was a prerequisite for the emergence of tax-levying elites and, eventually, states. We challenge this theory and propose that hierarchy arose as a result of the shift to dependence on appropriable cereal grains. Our empirical investigation, utilizing multiple data sets spanning several millennia, demonstrates a causal effect of the cultivation of cereals on hierarchy, without finding a similar effect for land productivity. We further support our claims with several case studies.

The conventional theory

- Farming raises land productivity.
- High output leaves food above the farmers' subsistence needs: a surplus.
- The surplus is the prerequisite for an elite, and for bureaucrats, troops and specialists who grow no food.
- So regional differences in land productivity should explain where hierarchy and states appear.
- The line runs from Adam Smith through Engels and Childe.

Appropriability instead

Mayshar, Moav and Pascali, The Origin of the State, JPE 2022

- “History records no cassava states, no sago, yam, taro, plantain, breadfruit, or sweet potato states” (Scott)
- Cereals are harvested in a short window and must be stored to last the year: durable, energy dense, easy to find and to carry.
- Roots and tubers are perennial, can be left in the ground, and rot soon after harvest. Around 70 percent water, so bulky to move.
- Stored grain is appropriable by the tax collector and by the bandit at the same time.
- So it creates both the opportunity to tax and the demand for protection.

Where the surplus comes from

- Hunter-gatherers also got more productive over time, and it went into population, not surplus.
- Their food was hand to mouth, so no elite could live off it.
- Once confiscation is possible, taxation shrinks the population, output per farmer rises, and a taxed surplus appears in equilibrium.
- The model: farmers split land between tubers, which cannot be looted or taxed, and cereals, which can.
- A state exists only where cereals are enough more productive than tubers to cover the fixed cost of deterring bandits.
- Hierarchy beats anarchy for farmers: the state can commit to a rate, and hires fewer collectors than anarchy breeds bandits.

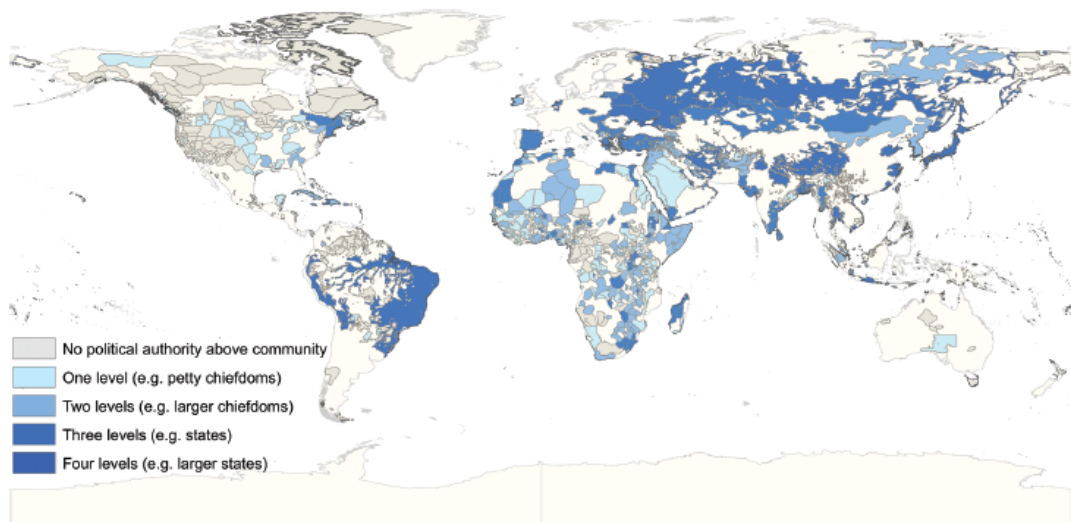


FIG. 1.—Jurisdictional hierarchy beyond the local community in preindustrial societies.

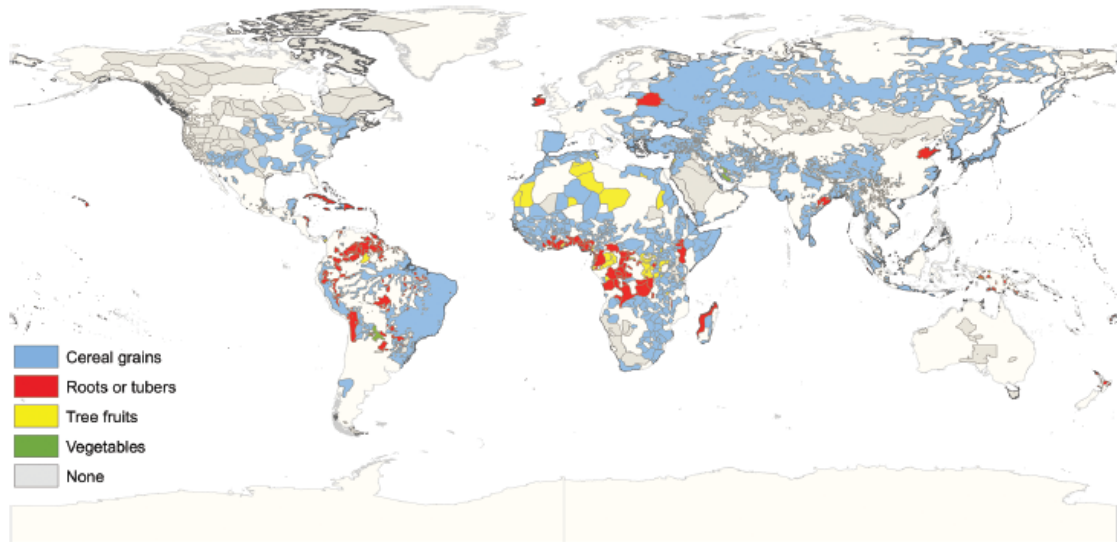


FIG. 2.—Major crops in preindustrial societies.

Hierarchy and crops, side by side

- Murdock's Ethnographic Atlas: 1,267 preindustrial societies, from before any significant industrialization.
- Jurisdictional hierarchy beyond the local community, five levels: none, petty chiefdoms, larger chiefdoms, states, large states.
- Most of the sample has no political integration above the local community.
- Second map: the main crop. About one fifth of the societies farm nothing at all.
- Compare them: the cereal band and the hierarchy band largely coincide. Root and tuber areas do not.

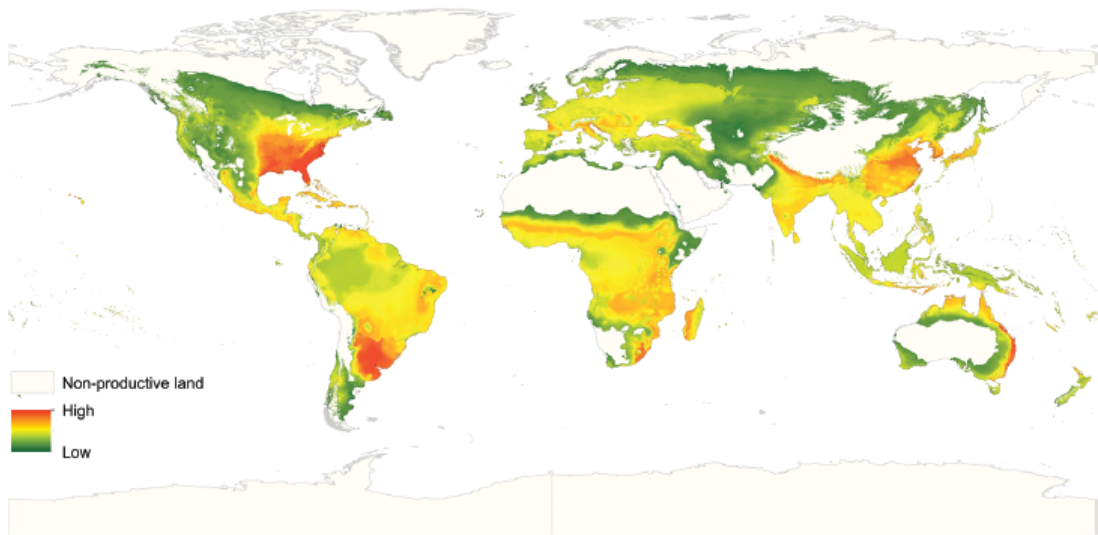


FIG. 3.—Difference in potential yields (calories per hectare) of cereals versus roots and tubers.

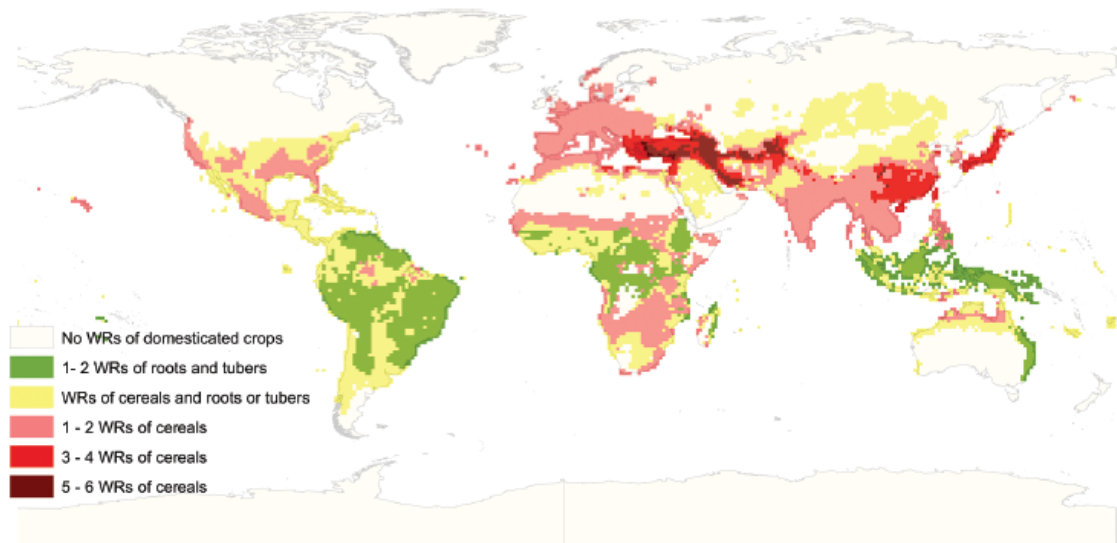


FIG. 4.—Distribution of wild relatives (WRs) of domesticated cereals, roots, and tubers.

Measuring what could have been grown

- First map: potential calories per hectare from cereals minus roots and tubers, under rain-fed subsistence farming, from agronomic suitability data.
- **Potential**, not actual: it does not respond to the institutions it is meant to explain.
- Second map: wild relatives of domesticated crops, built with the Global Crop Diversity Trust.
- More wild relatives is also Diamond's measure of a productive environment, so the two theories can be told apart on the same map.
- The combination that matters: cereals available, and no root or tuber alternative.

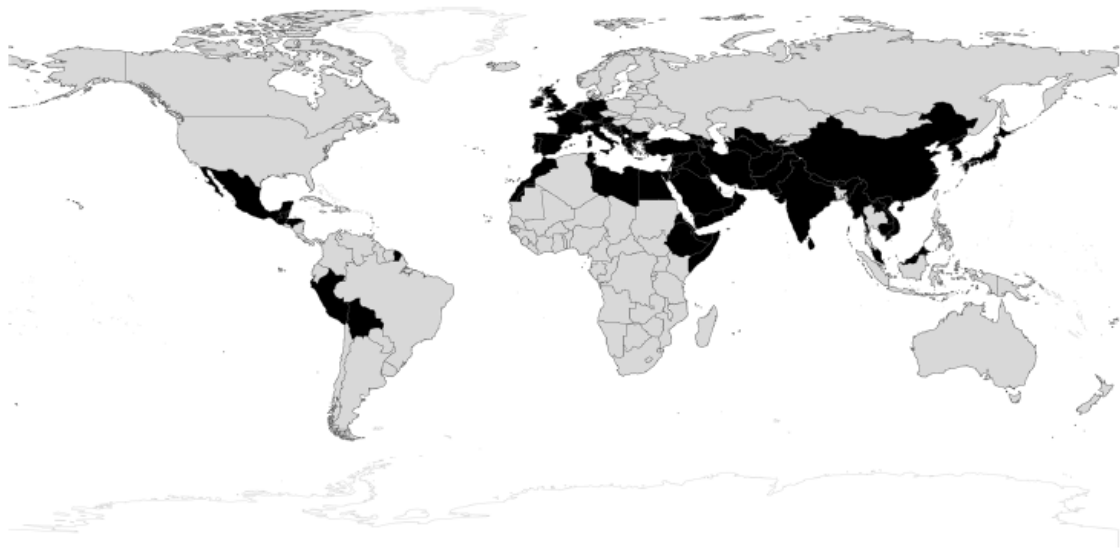


FIG. 5.—Modern countries' areas that were mainly organized as states in 450 CE.

States in 450 CE

- Areas mainly organized as states just before the collapse of the Roman Empire.
- Where only wild relatives of cereals were available, the hierarchy score is 0.50 higher, against a sample average of 0.38.
- That one contrast accounts for close to a third of the differences across regions.
- Hold the cereal advantage fixed and land productivity stops predicting hierarchy.
- Where roots or tubers were also available, hierarchy looks like the areas with no domesticable crops at all.

The rest of the evidence

- Preindustrial societies: land suited to cereals gets cereals, and cereal societies show deeper hierarchy and heavier taxation.
- Societies living on roots and tubers look like societies with no farming at all.
- The Columbian Exchange moved both land productivity and the cereal advantage. Only the cereal advantage moved hierarchy.
- Carbon-dated sites before and after the Neolithic: more traces of hierarchy afterwards, only where farming was likely to begin with cereals.
- None of the tests is decisive alone. The claim rests on their agreement.

Cases

- New Guinea farmed from about 5000 to 4500 BCE on bananas, taro and yam, with no complex hierarchy. The sweet potato arrived in the seventeenth century: output, population and feasting rose, hierarchy did not.
- Congo: a Bushong leader promoted maize and turned a tribal society into a kingdom. The Lele across the river avoided cereals, and avoided subjection.
- Jordan Valley, about 9000 BCE: an elaborate circular communal storage pit, before domestication. Communal storage needs someone to run it.
- Malawi today: women plant bitter, toxic cassava that takes far more work to prepare, because thieves prefer the sweet kind.

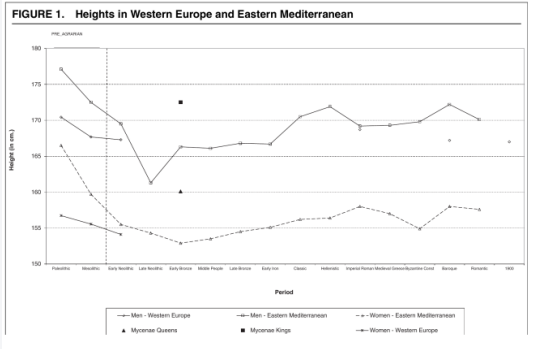
Latin America in this frame

- Amazonia: productive farming on manioc, and no complex hierarchy.
- The Andes: maize, plus potatoes freeze-dried in the mountains, and the Inca state.
- So elevation runs backwards here relative to Southeast Asia, where states sat in the rice valleys and the free peoples in the hills.
- In Matranga's account, South America is also a rainfall case, not a temperature case.
- Worth asking of Brazil: what was storable, who could reach it, and how far away could they carry it?

Boix and Rosenbluth (2014)

- *Bones of Contention: The Political Economy of Height Inequality*, APSR 2014.
- Bones as evidence on distribution: two people equally tall under good conditions end up different if one was undernourished while growing.
- Measure: the **coefficient of variation** of adult height inside a population, which is comparable across samples with different means.
- Claim: production technology and military technology open the potential for inequality, and institutions decide how much of it is realized.
- “we find that the shift from hunting and gathering to complex fishing techniques and to labor-intensive agriculture opened up inequalities that had discernible effects on human health and stature”

Mean height in the long run



- Men and women, Western Europe and the Eastern Mediterranean, Paleolithic to 1900.
- Pre-agrarian femurs suggest about 175 cm for men and 165 cm for women, within today's range.
- The paper's move: from the mean to the spread inside a population.

Native Americans in the Boas dataset

FIGURE 2. Native American Men: Average Height and Coefficient of Variation

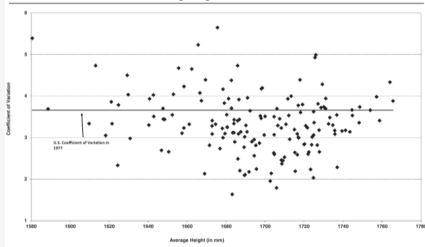
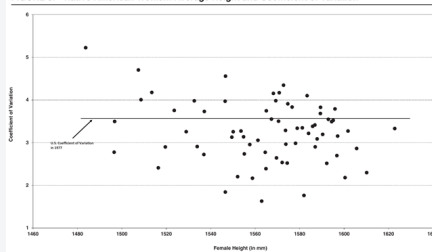


FIGURE 3. Native American Women: Average Height and Coefficient of Variation



- Franz Boas measured about 16,000 Native Americans from about 290 tribes in the late nineteenth century; averages 168.8 cm for men, 155.9 cm for women.
- Points: tribes with more than 10 observations. Horizontal line: the US value in 1977, 3.66 for boys and 3.57 for girls.
- About three quarters of tribes fall below that line, and around half are at 3.3 or lower.
- Simple production and simple warfare go with narrow height distributions.

Agrarian societies

FIGURE 5A. Distribution of Male Heights in Agrarian Economies

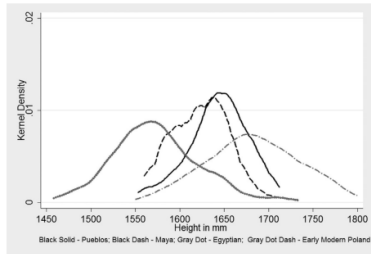
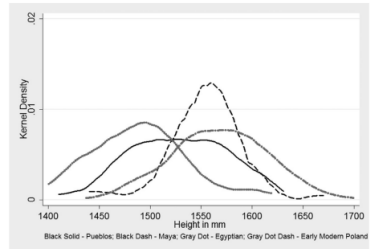


FIGURE 5B. Distribution of Female Heights in Agrarian Economies



- Zuni Pueblos and Maya: farming with rudimentary war technology. Egypt and early modern Poland: bronze, chariots, heavy cavalry, a military elite.
- Male height is far more dispersed in the second pair. So the split is not farming against foraging.
- Mycenae, measured directly: men in kings' tombs 172.5 cm, commoners 166.1 cm, a gap of 6.4 cm.

Height and dispersion since 1800

FIGURE 7A. Evolution of Male Heights and Coefficient of Variations since 1800

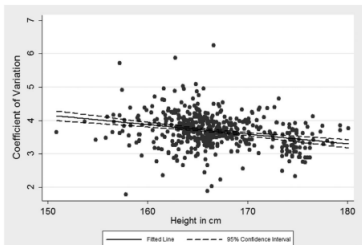
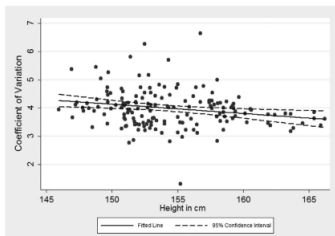


FIGURE 7B. Evolution of Female Heights and Coefficient of Variations since 1800.



Sources. The observations correspond to measurements of populations reported in the scientific publications employed by van Zanden et al. (2013).

- Each point is one population measured in the studies collected by van Zanden and coauthors.
- Mean height stands in for development.
- The coefficient of variation falls by about 0.75 points across the observed range of heights, for men and for women.
- Shorter populations are also more varied in how unequal they are.

Putting the four together

- Matranga: an uneven year pushed bands to settle and store, and farming followed.
- Mayshar, Moav and Pascali: what gets stored decides whether an elite can live off it.
- Olson: once appropriation is worth organizing, the bandit prefers to settle down and tax.
- Boix and Rosenbluth: the bones record how unequally the result was shared, and institutions move that.
- The recurring variable is storability and reach, not output.

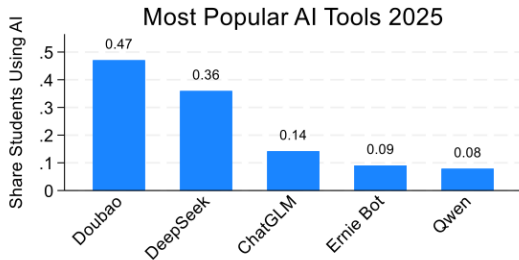
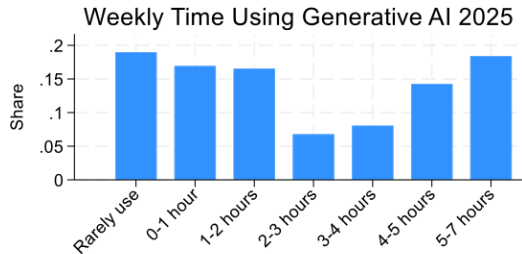
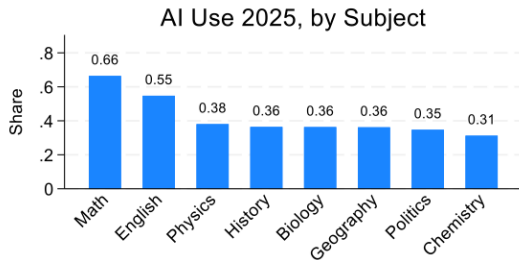
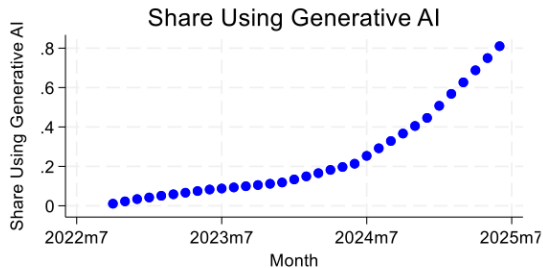
The Generative AI Learning Penalty: Evidence from Chinese Secondary Education*

David Strömberg[†] Victor Lei[‡] Yanhui Wu[§]

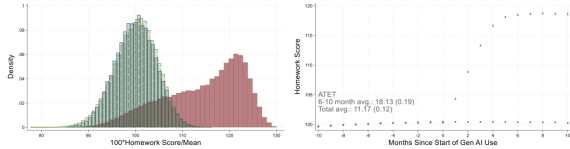
June 2026

Abstract

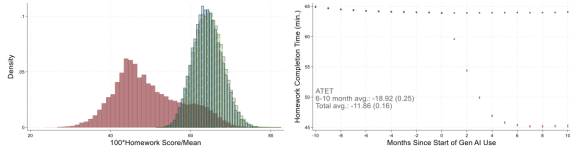
Using 30 months of panel data on 26,811 Chinese students in grades 7–12, we study how generative AI affects homework productivity and learning. The data combine monthly closed-book exams, high-school and college entrance exams, and homework scores and completion time across nine subjects. We exploit staggered AI adoption in a difference-in-differences design. AI adoption raises homework scores by 18% and reduces completion time by 30%, but lowers monthly exam scores by 20% within six months. High-stakes entrance-exam scores fall by 18 and 24%, with the full penalty emerging only after about two years. The losses are largest in social science subjects, followed by STEM and languages, and are especially large for junior students, high-achieving students, and boys. The learning losses are concentrated among roughly 80% of AI users whose behavior is consistent with homework outsourcing, as indicated by exceptionally short homework completion time coupled with high homework scores. AI users who maintain similar homework completion time as non-AI users experience small learning losses.



Panel A: Homework Score



Panel B: Homework Completion Time (min.)



Panel C: Exam Score

